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Implementation and performance of the Neural-Ringer electron selection algorithm in the high-level trigger of the ATLAS experiment

Electron trigger identification is a crucial input to many ATLAS physics analysis. To cope with the increase in peak LHC luminosity from 2017 to 2018, which raised the number of proton–proton interactions per beam-crossing to about 60, trigger algorithms were optimized to keep the background trigger rates low while retaining high efficiency for physics analyses. In the second half of 2017, the Neural-Ringer algorithm, based on a set of multi-layer perceptron models, replaced the cut-based strategy used in the FastCalo decision step in the high-level trigger and became the baseline algorithm for triggering events containing electrons with transverse energy above 15 GeV. This was done as part of the ATLAS online system CPU-time reduction campaign. It was the first time a neural-network method was employed as a baseline algorithm for event selection in the ATLAS trigger system, and resulted in the early rejection of at least 13 times more background than the cut-based strategy, while leaving the electron selection efficiency nearly unchanged.

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Implementation and performance of the Neural-Ringer electron selection algorithm in the high-level trigger of the ATLAS experiment

The ATLAS Collaboration

Electron trigger identification is a crucial input to many ATLAS physics analysis. To cope with the increase in peak LHC luminosity from 2017 to 2018, which raised the number of proton–proton interactions per beam-crossing to about 60, trigger algorithms were optimized to keep the background trigger rates low while retaining high efficiency for physics analyses. In the second half of 2017, the Neural-Ringer algorithm, based on a set of multi-layer perceptron models, replaced the cut-based strategy used in the FastCalo decision step in the high-level trigger and became the baseline algorithm for triggering events containing electrons with transverse energy above 15 GeV. This was done as part of the ATLAS online system CPU-time reduction campaign. It was the first time a neural-network method was employed as a baseline algorithm for event selection in the ATLAS trigger system, and resulted in the early rejection of at least 13 times more background than the cut-based strategy, while leaving the electron selection efficiency nearly unchanged.

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Contents

1 Introduction

The ATLAS physics programme relies on an efficient trigger system [1] to record a highly signal-enriched subset of collision events produced by the Large Hadron Collider (LHC) at CERN. Electron triggers [2] are important for searches for new particles, measurements of Standard Model cross-sections, and precise measurements of the properties of fundamental particles such as the Higgs [3, 4] and W bosons.

A hierarchical two-level trigger system is used to select events of interest produced in LHC beam collisions. The first-level (L1) trigger, implemented with custom-made electronics located on the detector, utilizes the calorimeters at coarser granularity to reduce the event rate from the 40 MHz bunch-crossing rate to below 100 kHz, of which about 35% corresponds to events with high-energy electromagnetic showers, potentially coming from electrons. The L1 trigger also defines regions-of-interest (RoIs) [5], situated around local regions with high transverse energy.

The RoIs defined by the L1 trigger are processed by the high-level trigger (HLT) implemented in software. The HLT operates in a large computing farm and should reduce the rate of events written to disk to an average of about 1 kHz. For electron selection, the HLT is split into two stages: a fast but efficient computation followed by precise processing, where the latter employs offline-inspired algorithms adapted to the harsh conditions (e.g. time and memory constraints) of the trigger system.

The schematic diagram in Figure 1 shows the chain of algorithms for electron reconstruction in the HLT. Electron selection starts with a fast calorimeter-reconstruction step (FastCalo), which has two implementations: the original cut-based selection and the new neural-network (NN) approach using the Neural-Ringer algorithm presented in this paper. This is followed by a fast track-reconstruction step (FastElectron), which also places requirements on variables that indicate the quality of proximity matching between the electron candidate positions reconstructed from the track and the calorimetric measurements. The HLT tries to select particle candidates at each step; if it fails at a certain step, subsequent steps are not executed. This is essential in order to reduce the CPU resources and time needed by the HLT to reconstruct the event and make a trigger decision.

This paper describes the performance improvements due to the introduction of the Neural-Ringer algorithm into the HLT fast calorimeter-reconstruction step for electron triggers. It became the baseline algorithm for selecting events containing at least one isolated electron with transverse energy above 15 GeV in the 2017 proton–proton collision ATLAS data-taking at $\sqrt{s}$ = 13 TeV in LHC Run 2. Full details of its electron identification procedure, including the training and tuning strategies, are given in Section 2. Online performance results from the Neural-Ringer and cut-based strategies are presented in Section 3. The Neural-Ringer and cut-based electron triggers are compared from an offline perspective using statistical methods in Section 4. Finally, the conclusions and prospects are presented in Section 5.

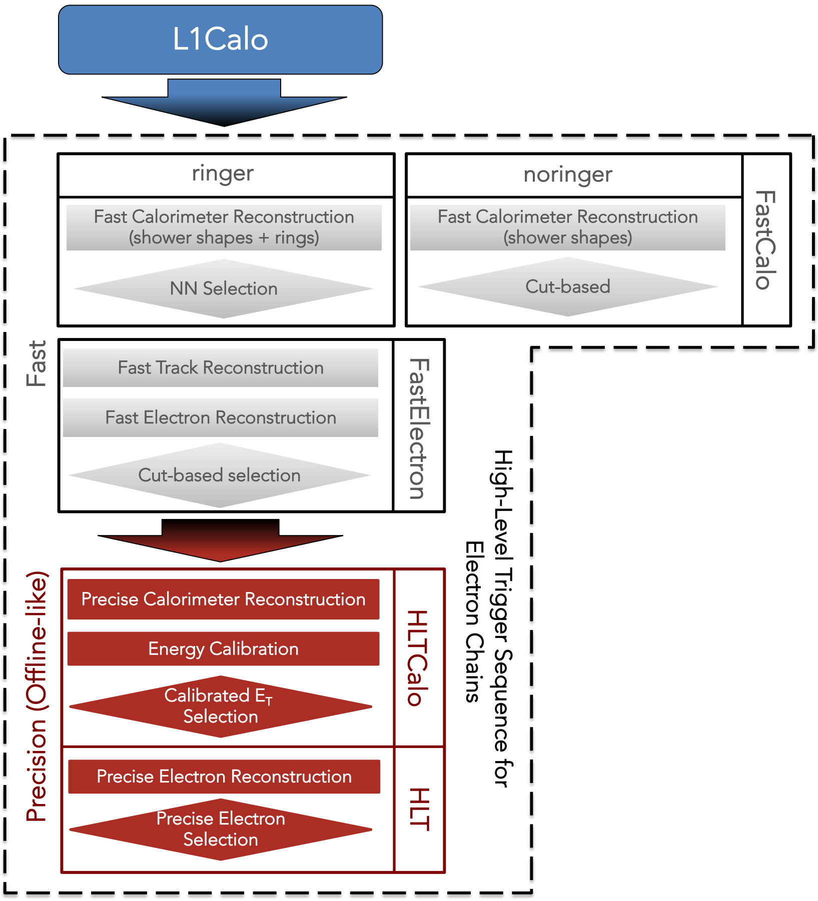


Figure 1: Chain of algorithms for electron triggers in the HLT. Differences between the original baseline (no-ringer) and new (ringer) electron triggers lie in the first step, called FastCalo. Grey blocks represent algorithms in the fast stage, and red blocks represent precision-step algorithms.

2 The Neural-Ringer algorithm

Instead of using electron shower-shape variables [1] to discriminate between electrons and background, the Neural-Ringer algorithm adopts a shower representation consisting of rings of transverse energy, built from sums of calorimeter-cell energies, around each candidate provided as an RoI by the L1 trigger. The distribution of those energy sums is used by a set of neural networks to distinguish electrons from background. This section describes the Neural-Ringer algorithm.

2.1 The ATLAS calorimeters

The ATLAS calorimeters comprise rectangular cells in the $\eta \times \phi$ plane[1] for different layers in depth, except for the forward calorimeters. The calorimeter system covers the pseudorapidity range $|\eta| < 4.9$ [6]. Within the region $|\eta| < 3.2$, electromagnetic calorimetry is provided by a high-granularity lead/liquid-argon (LAr) calorimeter (ECal) composed of barrel (EMB1,2,3) sections covering $|\eta| < 1.475$ and endcap (EMEC1,2,3) sections covering $1.375 < |\eta| < 3.2$, with an additional thin LAr presampler (PS) covering the barrel (PreSamplerB) and endcap (PreSamplerE) regions to correct for energy losses in upstream material [7, 8]. Hadronic calorimetry within $|\eta| < 3.2$ is provided by a steel/scintillator-tile calorimeter (TileCal) [9, 10] composed of three barrel sections (TileBar0,1,2) covering $|\eta| < 1.0$ and three extended barrel sections (TileExt1,2,3) covering $0.8 < |\eta| < 1.7$, and two copper/LAr hadronic endcap calorimeters (HEC) [11] divided into four modules (HEC0,1,2,3) covering $1.5 < |\eta| < 3.2$. Transition regions between calorimeters are used as locations for detector services and induce shower-sharing between calorimeters, which degrades energy measurements in those regions. Here, the barrel-to-endcap transition is complemented by the intermediate tile calorimeter (ITC), which is composed of two modules (TileGap1,2,3) that are attached to the external faces of the barrel and are used to estimate signal losses [11]. The solid-angle coverage is completed with forward copper/LAr and tungsten/LAr calorimeter (FCal) modules optimized for electromagnetic (EM) and hadronic (HAD) measurements, respectively. The calorimeter system provides full azimuthal (ϕ) coverage with a total of about 190 000 readout cells.

The electromagnetic and hadronic calorimeters are segmented [6] longitudinally (i.e. in depth) into three[2] sampling layers, where each layer has its own lateral ($\eta \times \phi$) granularity. The amount of material in front of each sampling layer varies with $|\eta|$, leading to variations in the expected lateral and longitudinal shower profiles. The expected profiles also depend on the physics object's total energy. In the EM calorimeter, most of the EM shower energy is collected in the second layer (EM2), while the third layer (EM3) provides measurements of energy deposited by the shower tails. However, it should be noted that the first EM layer (EM1) plays an important role in discriminating between electrons and π^0 mesons. The hadronic calorimeters, which surround the EM detectors, provide additional electron discrimination power through energy measurements of possible EM shower tails, as well as rejection of events with activity of hadronic origin via the three sampling layers (HAD1,2,3).

[1] ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z-axis along the beam-pipe. The x-axis points from the IP to the centre of the LHC ring, and the y-axis points upward. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the beam-pipe. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$. The angular distance ΔR is defined as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. Transverse momenta and energies are defined as $p_{\mathrm{T}} = p \sin\theta$ and $E_{\mathrm{T}} = E \sin\theta$, respectively.

[2] The two central layers of each hadronic endcap calorimeter are summed to provide a single measurement.

2.2 Building the rings

Since the Neural-Ringer algorithm was designed to operate online in software at the FastCalo stage, approximations were made. Specifically, rectangular rings are constructed to match the grid layout of the calorimeter cells, and the ring coverage is bounded by the RoI's area. In addition, granularity changes due to different cell sizes in different regions of the detector are neglected in order to employ a grid-like algorithm. When considering only the EM1 layer, the η-granularity used by the algorithm is 0.003 for the entire η range defined ($|\eta| < 2.5$), although this layer actually covers the region $0 < |\eta| < 2.37$ and has three different η-granularities: 0.003 for $0 < |\eta| < 1.81$, 0.004 for $1.81 < |\eta| < 2.01$ and 0.006 at $2.01 < |\eta| < 2.37$.

The algorithm's RoI covers a calorimeter region of 0.4×0.4 in the $\eta \times \phi$ plane, centred on the L1-provided RoI. The algorithm starts at the second EM sampling layer (EM2), where the $\eta \times \phi$ position of the cell with the highest E_{T} in the algorithm's RoI is taken as the centre of the cascade interaction in the calorimeter. The initial seed-cell position is propagated to other calorimeter layers in order to define the axis of the rings in those layers. Then, for each layer, the algorithm refines the seed position by searching for the most energetic cell inside a reconstruction window centred on the EM2 initial seed, and the energy deposition of an incoming particle is extracted by building concentric rings of cells, simply referred to as 'rings'.

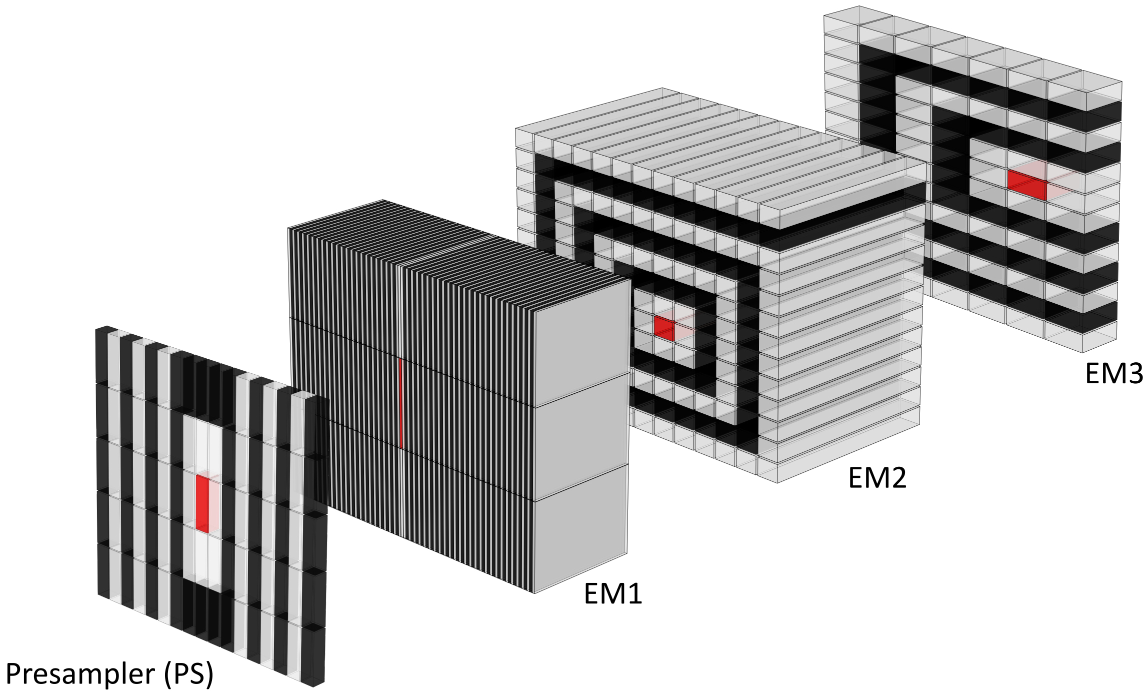
The refined seed ($c_{\mathrm{hot},l}$) is used as the starting point to build the rings. A ring $R_{n,l}$ contains all $C_{n,l}$ cells in calorimeter layer l which are n cells away from the refined seed (see Figure 2 for an illustration of the parameters). Formally,

$$R_{n,l} = \left\{ c_{n,l} \mid n = \left\lfloor \max\left(\frac{|\eta_{i,l} - \eta_{\mathrm{hot},l}|}{h_{\eta,l}}, \frac{|\phi_{i,l} - \phi_{\mathrm{hot},l}|}{h_{\phi,l}} \right) \right\rceil, \forall c_{i,l} \in \Theta_{\mathrm{RoI},l} \right\},$$

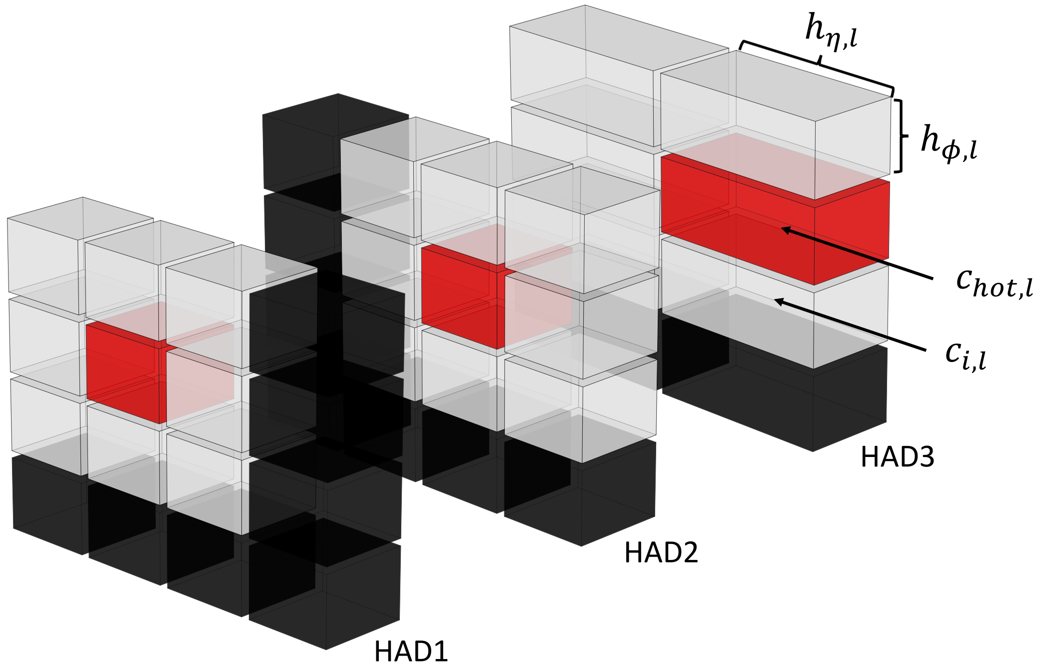
where (and analogously for ϕ when appropriate) $\eta_{i,l}$ and $\eta_{\mathrm{hot},l}$ are respectively the $c_{i,l}$ and $c_{\mathrm{hot},l}$ cells' positions in η;$h_{\eta,l}$ is the l^{th} layer's cell size in η; $\Theta_{\mathrm{RoI},l}$ is the set of cells in the l^{th} layer which are within the Ringer RoI; and $\lfloor \cdot \rceil$ is the *round* function. Table 1 shows the number of rings computed at each layer and the respective granularity.

Table 1: Nomenclature defining the Neural-Ringer algorithm layers and sections, as well as the respective parameters employed and the calorimeter sampling layers from which the cells are extracted. The Neural-Ringer algorithm parameters were dependent on the ringer layer l but independent of η and E_{T} during LHC Run 2. The parameters are the ring size in η ($h_{\eta,l}$), ϕ ($h_{\phi,l}$) and the number of rings to be computed in each layer (N_l).

Ringer		Calorimeter Sampling			Parameters		
Section	Layer (l)	Barrel	ITC	End-cap	$h_{\eta,l}$	$h_{\phi,l}$	N_l
EM	PS	PreSamplerB		PreSamplerE	0.025	0.1	8
	EM1	EMB1		EMEC1	0.003	0.1	64
	EM2	EMB2		EMEC2	0.025	0.025	8
	EM3	EMB3		EMEC3	0.050	0.025	8
HAD	HAD1	TileBar0 TileExt0	TileGap3	HEC0	0.1	0.1	4
	HAD2	TileBar1 TileExt1	TileGap1	HEC1 HEC2	0.1	0.1	4
	HAD3	TileBar2 TileExt2	TileGap2	HEC3	0.2	0.1	4



(a) Electromagnetic calorimeter cells within the ringer reconstruction window.



(b) Hadronic calorimeter cells within the ringer reconstruction window.

Figure 2: Sketch illustrating the ring-shaped energy description. The most energetic cell is in red, while the consecutive neighbouring rings are represented by alternating grey and black cells.

The sums of the transverse energies of cells $c_{n,l}$ in the rings $R_{n,l}$ form a vector of discriminating quantities. They are ordered laterally outwards and longitudinally from the innermost to outermost layer of the calorimeter, thus providing the ring-shaped information about the RoI.

The ring-building process results in a total of 100 rings across all calorimeter layers in each RoI. Well-known features of EM shower physics allow a large reduction in dimensionality to be achieved by compacting the typical input space of approximately 1000–1200 cells per ROI into these 100 rings: the rings can keep a complete description of the longitudinal and symmetric lateral information. However, the algorithm only approximates this concept in order to meet the online operation requirements and avoid further manipulation of the instrumental information. Figure 3 shows the ring-shaped mean profile differences between electrons and jet particles.

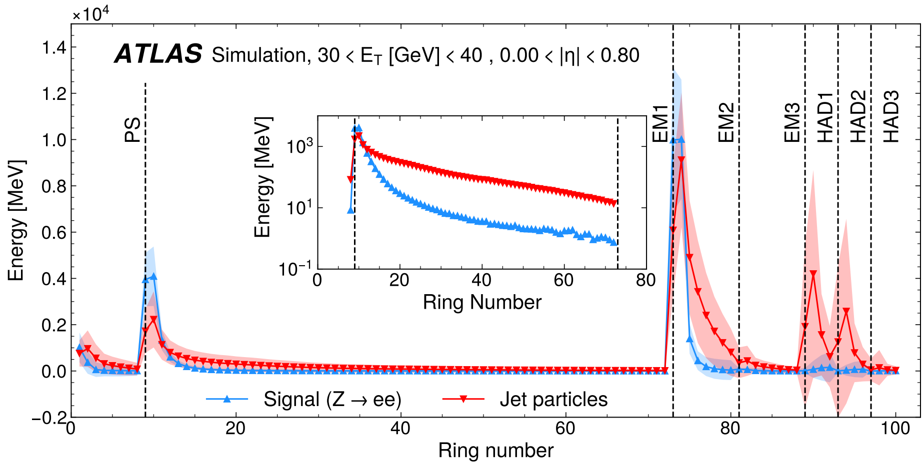


Figure 3: Average ring profile from simulation data at $30 \le E_{\mathrm{T}} < 40$ GeV and $0.0 \le \eta < 0.8$ for each layer of the calorimeter. The profile in the inset shows the EM1-layer ring energies on a logarithmic scale. Most of the energy of the electrons is absorbed by the electromagnetic layers. For the jets, most of the energy is absorbed by the hadronic layers.

2.2.1 Ring sums as discriminating variables

The current strategy concatenates all rings into a single vector of 100 variables. The variables are normalized to the absolute value of the total RoI energy, as shown in Eq. (1):

$$r'_k = \frac{r_k}{|\sum_{i=1}^{100} r_i|}, \quad \forall\, k \in \{1, \ldots, 100\}, \tag{1}$$

where r_k or r_i is the sum of transverse energy of the cells in the ring with index k or i, and r'_k is the normalized ring value with index k. The absolute value is used because of negative noise accumulation from some cells within the reconstruction window that were not excited by the particle shower. This procedure was initially proposed and examined in Ref. [12] as a way of preserving the shower energy profile as fractions of the total energy. It is especially valuable because of its simplicity, since it uses a non-parametric approach, and for allowing easy interpretation of the shower profile.

2.3 Motivation for using a set of neural networks

The normalized concatenated ring vector is fed into a set of multi-layer perceptrons (MLPs). Each MLP corresponds to a model specifically trained for a particular region of $E_{\mathrm{T}} \times |\eta|$ space. The model chosen for each shower's vector is the one trained for the nearest region of phase space.

For calorimetry, the use of a set of models, each specific to a particular phase-space region, mitigates fluctuations in the variables' profiles due to differences in shower energy and detector response. A large contribution comes from discrete detector-granularity changes in the phase space. Other important factors influencing such profiles are the amount of the detector material as a function of $|\eta|$ and the dependence of the underlying shower-development processes on the incoming particle's energy. Although the latter changes are mostly continuous, it is also possible to approach the problem via space discretization.

At the hardware resource level, splitting the training into multiple small models to accommodate detector response inhomogeneities speeds up the training cycle and allows data to be handled in memory, since only a small portion of the data is used to train a specific model. Limiting the memory required for tuning the models was of particular interest in order to benefit from low-memory processing nodes, as these were the ones mostly available in the Worldwide LHC Computing Grid [13] during our development effort. In addition, the decomposition of the problem can also lead to a reduction in training time [14], as observed heuristically when comparing the tuning time of the set with that of a single overall model. With regard to operation, the proposed set only requires the choice of a single model in order to operate, whereas the more common committee-machine approaches require the computation of different models to run in parallel with a fusion method [15]. Therefore, the chosen set configuration adds minimal overhead in terms of trigger latency. Similarly, our choice was to use low-complexity models, with a single hidden layer and few neurons, to develop a method competitive with the cut-based approach in terms of processing speed.

2.4 Training and decision making for 2017 and 2018 operation

The training procedure for 2017 operation was based on samples of simulated $Z \to ee$ decays, selected with a $Z \to ee$ tag-and-probe method [1], and background. For simulated background samples, a filter was applied to enrich them with particles produced in the hard scatter with a summed transverse energy exceeding 17 GeV in a region of $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$. During 2016 data-taking, pile-up reached 40 interactions per bunch crossing. Both types of simulated samples included an average of 60 pile-up interactions per bunch crossing, which was not seen in collision data until the end of 2017, when the peak pile-up exceeded 60. Recently, in 2023, the highest observed level of pile-up exceeded 60 again.

After the training stage, threshold values for the discriminating variables must be determined in order to select good electron candidates. To set the correct values without causing any HLT inefficiency for collision data, probe electrons selected by the $Z \to ee$ tag-and-probe method from the full 13 TeV pp collision dataset recorded by ATLAS in 2016 were used if they also satisfied the tightest available offline criteria. For background, objects rejected by the $Z \to ee$ tag-and-probe method, which do not belong to any tag–probe electron pair, were used if they were also rejected by the loosest available offline criteria. For 2018 operation, a purely data-driven strategy was used, selecting events from the 2017 recorded collision data to train the models and adjust all thresholds. To be used for tuning and analysis, the recorded collision data were first required to pass the standard ATLAS data quality procedure.

2.4.1 Training and decision making

The 2017 procedure derived a specific MLP model for each region of $E_{\mathrm{T}} \times |\eta|$ phase space, using simulated samples, and defined threshold values for the discriminating variables by using 2016 collision data and targeting a signal efficiency similar to that of the cut-based trigger. The training procedure and decision-making processes were the same for all phase-space regions. For the Neural-Ringer, each MLP is an expert model for a single phase-space region, containing its own topology and parameters.

The model parameters were optimized using events selected from simulation datasets. The structure consists of a single fully-connected hidden layer, which may contain from 5 to 20 hidden units, an input layer with 100 inputs, one for each ring, and an output layer with a single neuron. The activation function for both the hidden and output neurons is a hyperbolic tangent, which has often been used in MLP designs [16].

To assess the statistical fluctuations of the efficiency measurements, the stratified k-fold cross-validation method [16], with $k = 10$, was used, although repeating the training and testing procedures on different randomly chosen subsets increased the computational cost. The stratified k-fold is one of the most common cross-validation techniques and consists of partitioning the dataset into k disjoint subsets, each model being tested with the i^{th} subset after being trained with the other ones. The dataset stratification follows the empirical order of the event selection, in order to allow straightforward job parallelization, i.e. random permutation is not employed.

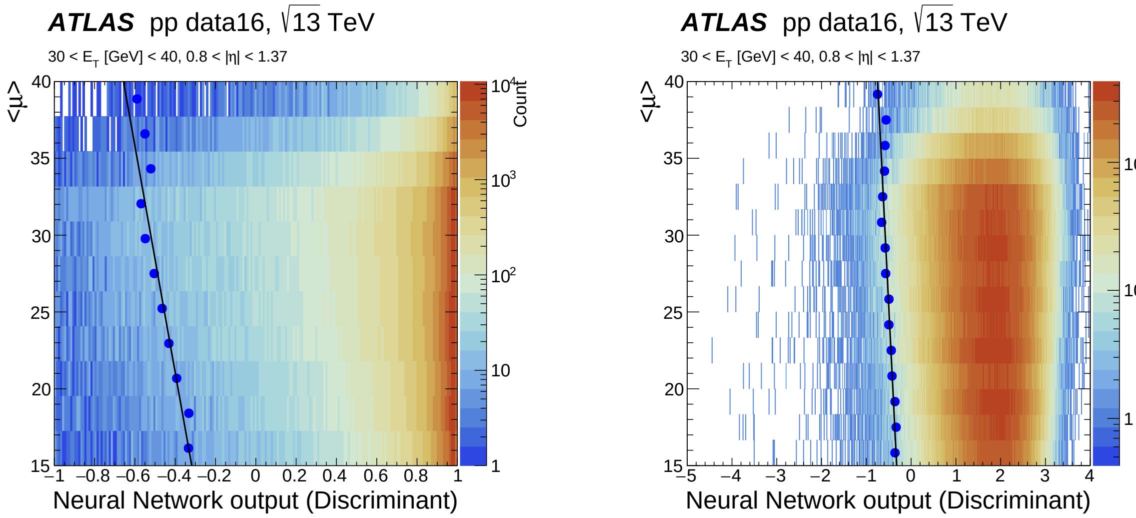
For each cross-validation sort and working point, 100 networks, initialized as in Ref. [17], are optimized by back-propagating the mean squared error through the RPROP algorithm [18]. This repeated model initialization seeks to avoid poor sub-optimal solutions due to the use of gradient-based algorithms. For each training procedure, only three models out of the 100 are retained: two for retrieving the best efficiency when fixing the sensitivity or false positive probabilities to the baseline FastCalo efficiency and another that gives the $\mathrm{SP}_{\mathrm{max}}$ value. The SP value is defined as:

$$\sqrt{\sqrt{P_{\mathrm{D}}(1 - F_{\mathrm{A}})} \cdot \frac{P_{\mathrm{D}} + (1 - F_{\mathrm{A}})}{2}}$$

where P_{D} is the signal detection probability (or sensitivity) and F_{A} is the probability of a fake signal (false positive). An early stopping [16] technique is employed to avoid model over-training. The selection of the optimal iteration and operating model is based on a heuristic approach (multi-stop) [19].

Computational resources were used to tune 1.3 M shallow-learning neural networks, which resulted in a set of 20 MLPs (five in E_{T} × four in $|\eta|$) for each one of the four working points used by electron triggers in the HLT. Thus, the first training strategy considered only four regions in $|\eta|$ ($0.0 \to 0.8 \to 1.37 \to 2.5$) and five regions in E_{T} (15 GeV → 20 GeV → 30 GeV → 40 GeV → 50 GeV → ∞). Apart from few exceptions, the models in the set employed five neurons in the hidden layer.

After the training stage, decisions are made by applying a threshold requirement to the one-dimensional output node of each expert MLP. As in the HLT likelihood algorithm, the threshold value used in the Neural-Ringer algorithm is computed as a linear function of $\langle\mu\rangle$, the mean number of pile-up interactions per bunch crossing. However, non-linear behaviour was observed near the asymptote of the output activation function (see Figure 4(a)) when using the 'medium' or 'tight' operating point. Heuristically, it was observed that this behaviour can be avoided by replacing the hyperbolic tangent activation function in the output neuron with a linear function (Figure 4(b)) for operation, after the training stage is completed. This strategy ensures linear behaviour of the targeted signal efficiency with respect to the online pile-up estimator. To account for Monte Carlo and collision data mis-modelling (see Figure 5), the parameters of the linear threshold correction were derived using 2016 collision data and an upper bound of $\langle\mu\rangle = 40$ pile-up interactions (i.e. the corrected threshold value is constant for pile-up values above the upper bound).



(a) MLP output with a hyperbolic tangent activation function in the output neuron versus pile-up.

(b) MLP output with a linear activation function in the output neuron versus pile-up.

Figure 4: Neural-Ringer output as a function of pile-up, $\langle\mu\rangle$, for $Z \to ee$ probes in 2016 collision data selected using the training criteria. The computed threshold per $\langle\mu\rangle$ unit for achieving the targeted efficiency is shown as a full blue circle. The black line is a linear fit to the thresholds. The output space is computed using the trained model without modifications in (a), while in (b) the activation function in the output neuron is replaced with a linear function.

It was observed that the rings in the region $2.37 < |\eta| < 2.47$ had particular profiles due to the lack of strip cells in this region and required a specific discriminant threshold in order to keep the signal efficiencies close to those of the cut-based triggers. In order to avoid retraining all models, it was decided to split the last $|\eta|$ region in two, from $1.37 \to 2.5$ to $1.37 \to 2.37 \to 2.5$, and duplicate all models, for this region, to the new bin. Finally, the set used to operate in 2017, which combines every η boundary with each E_{T} boundary, has a total of 25 MLP models. The set's boundaries are defined in Table 2.

Table 2: E_{T} and η boundaries used to create the set of neural networks.

Model Adjustment	
E_{T} [GeV] boundaries	$|\eta|$ boundaries
$15 \leq E_{\mathrm{T}} < 20$	$0.00 \leq |\eta| < 0.80$
$20 \leq E_{\mathrm{T}} < 30$	$0.80 \leq |\eta| < 1.37$
$30 \leq E_{\mathrm{T}} < 40$	$1.37 \leq |\eta| < 1.54$
$40 \leq E_{\mathrm{T}} < 50$	$1.54 \leq |\eta| < 2.37$
$E_{\mathrm{T}} \geq 50$	$2.37 \leq |\eta| < 2.47$

On the other hand, the 2018 training procedure employed a single set structure for all working points and used the 2017 collision data, selected with $Z \to ee$ tag-and-probe event selection, for training, as was the case for the derivation of the offline and final HLT likelihood models [2]. Most of the procedure was kept unchanged for 2018. The developments for 2018 data benefited from the data conditions being similar

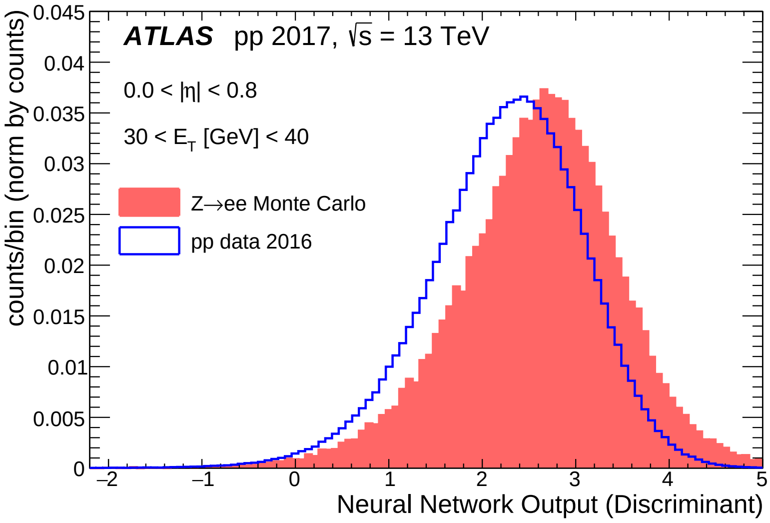


Figure 5: Differences between simulation samples (red) and collision data (blue) obtained in 2016 for one region of the phase space. The comparison is made between the normalized distributions of the output of the neural network (the discriminant) obtained for these two cases. The shift in discriminant value between simulation and collision data is caused by differences between the simulation and data rings used as input to the model.

to those in 2017. To simplify the training strategy, the selection of three models for each training was abandoned, keeping only the SP_{max} model, as the previous approach was more complex without a clear increase in trigger efficiency.

Likewise, MLPs with 5 to 10 units in the hidden layer were tuned, as most models did not require more than 10 units in 2017. The training optimizer was changed to *Adam* [20], instead of RPROP, and the MLP model topology was modified to have a hidden layer with fully connected neurons, with rectified linear units (RELU) as the activation function, and an output layer with one neuron using a sigmoid activation function. With a larger time span for entering operation, specialized MLPs were implemented in the region $2.37 < |\eta| < 2.47$, where different ring profiles were observed before 2017 operation, resulting in the operation of 25 MLPs. Finally, the upper bound for threshold corrections was set to $\langle\mu\rangle = 100$ pile-up interactions in order to extrapolate the operation of the models to a never-reached pile-up level.

3 Online deployment and performance

The Neural-Ringer algorithm was implemented in four stages during 2017–2018 in LHC Run 2:

i A development stage up to early 2017, where the Neural-Ringer's potential was estimated from trigger emulation and data reprocessing.

ii The commissioning stage in early 2017 runs (5.4 fb^{-1}), where all primary triggers were duplicated and either the Neural-Ringer or cut-based algorithm operated in the FastCalo stage of the HLT.

iii The operation stage, starting after Technical Stop 1 (TS1) in 2017. Triggers employing the Neural-Ringer were used as the main triggers, and cut-based triggers were kept as backup triggers. Final validation of the Neural-Ringer was performed by comparing data from this period (39.0 fb^{-1}) as shown in Section 3.1 and Section 4.

iv The Neural-Ringer-only stage, during 2018. Duplicated cut-based triggers were removed. The performance of the Neural-Ringer in 2018 is presented in Section 3.2.

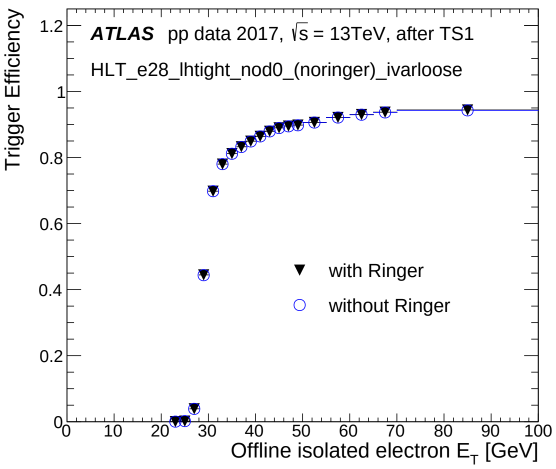
This section reports the Neural-Ringer's operation in each of these periods. In addition, CPU-time usage is discussed in Section 3.3.

3.1 2017 operation

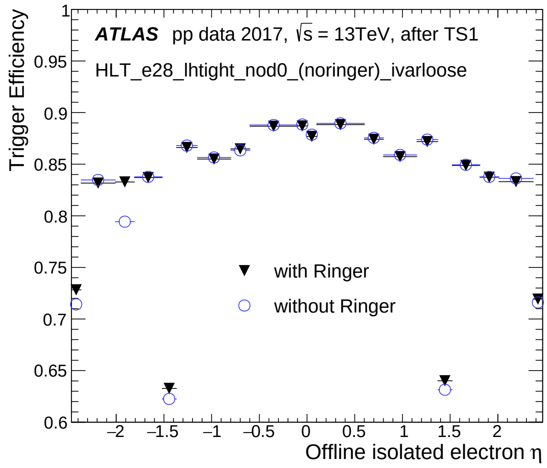
A backup single-electron HLT_e28_lhtight_nod0_(noringer)_ivarloose trigger, both with and without (noringer) the Neural-Ringer, was employed for monitoring purposes after the first technical stop (TS1). This trigger requires an electron candidate with $E_{\text{T}} > 28$ GeV satisfying the 'tight' selection (lhtight_nod0) with 'loose' isolation criteria (ivarloose) [1]. During the monitoring process, the two trigger variants were observed to operate similarly in terms of signal efficiency for the integrated luminosity during the period, as shown in Figure 6. The trigger turn-on curves exhibit similar E_{T} profiles. The Neural-Ringer shows a reasonably symmetric profile for positive and negative η values. In addition, an efficiency difference of about 0.5% to 1‰ can be observed in the transition region near $|\eta| = 1.5$ between the endcaps and barrel. Apart from these features, the overall efficiency differences are typically smaller than a few per mille. Although a slightly more prominent efficiency loss is observed at large $\langle\mu\rangle$ in Figure 6(c) for the Neural-Ringer trigger, the electron efficiency was kept nearly the same. This loss occurs after the linear threshold correction limit of $\langle\mu\rangle = 40$ was employed during 2017.

Other important triggers were assessed by comparing the trigger efficiencies in 2017 collision data collected before and after switching to the Neural-Ringer algorithms, as shown in Figure 7. The single-electron HLT_e17_lhvloose_nod0_L1EM15VHI trigger requires an electron candidate with $E_{\text{T}} > 17$ GeV, seeded by the first-level trigger L1_EM15VHI, and satisfying the loosest identification criteria (lhvloose). On the other hand, HLT_e26_lhtight_nod0_ivarloose requires an electron candidate with $E_{\text{T}} > 26$ GeV satisfying 'tight' identification (lhtight) with 'loose' isolation (ivarloose), and HLT_e60_lhmedium_nod0 requires an electron candidate with $E_{\text{T}} > 60$ GeV satisfying 'medium' identification (lhmedium). The electron efficiency was also kept nearly unchanged for these relevant single-electron triggers. As mentioned above, the results in Figure 7 are computed for two different data-taking periods: one with, and the other without, the Neural-Ringer algorithm.

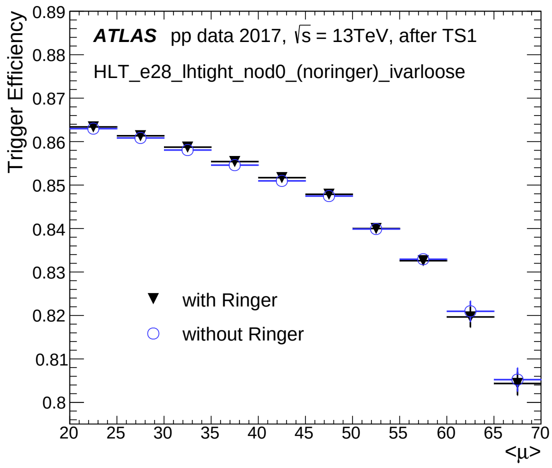
The power of the Neural-Ringer algorithm is made clear by examining the behaviour of the duplicated trigger for fake electron candidates in collision data. The overall fake-electron rate is reduced by a factor of



(a)



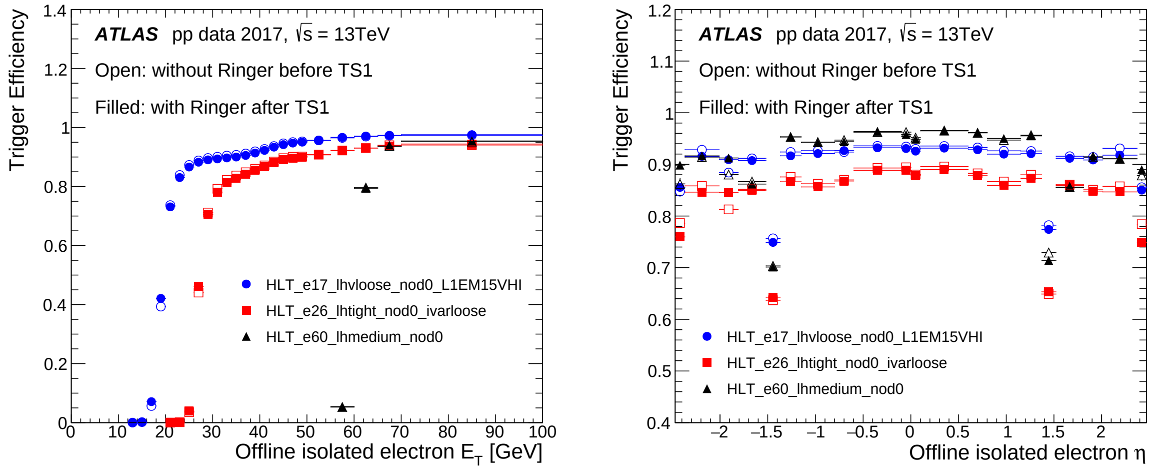
(b)



(c)

Figure 6: HLT electron efficiency in 2017 collision data as a function of (a) E_{T}, (b) η, and (c) $\langle\mu\rangle$ for the single-isolated-electron trigger requiring $E_{\mathrm{T}} > 28$ GeV and 'tight' selection with and without the Neural-Ringer algorithm after its deployment.

13.75 at the FastCalo step. It can be seen on the left side of Figure 8 that the improvement at the FastCalo step is similar for all regions in the evaluated variables, which is particularly interesting when considering the low E_{T} and endcap regions. Besides improving early fake-electron rejection, use of the Neural-Ringer also resulted in a factor of two reduction in the final fake-electron rate from the HLT (see right side of Figure 8), mostly coming from the calorimeter's barrel–endcap transition regions.



(a) (b)

Figure 7: Efficiency of three single-electron triggers as a function of (a) E_{T} and (b) η for 2017 collision data. Open (filled) markers show the efficiency measurements in runs before (after) the deployment of the Neural-Ringer, thus referring to triggers executed without (with) the Neural-Ringer algorithm.

3.2 2018 operation

For 2018 operation, the Neural-Ringer was retuned using collision data. The efficiency of the lowest-transverse-energy-threshold unprescaled trigger increased by at least 1%, due to better operation in all selection steps, and particularly from improvements in the likelihood tunes.

As well as maintaining high electron efficiency, some unprescaled triggers were observed to achieve a small reduction in fake-electron acceptance at the FastCalo step in comparison with 2017 operation after TS1. For the lowest-E_{T}-threshold trigger, the fake-electron acceptance was reduced by a factor of 1.19, while for the highest-E_{T}-threshold trigger, a reduction factor of 1.69 was achieved.

3.3 Impact on CPU demands

As observed in the efficiency measurements during 2017–2018 data-taking, the Neural-Ringer allowed more effective FastCalo operation than with the previous cut-based approach for triggers requiring electron E_{T} above 15 GeV.

The overhead required to compute the NN decision was evaluated by running a single-electron trigger with and without the Neural-Ringer algorithm, in two individual non-concurrent executions using the same dedicated node. It should be mentioned that, in order to reduce implementation effort, electron triggers using Neural-Ringer information compute their decision after the cut-based reconstruction, for which a trigger decision is not computed. As a result of this, the Neural-Ringer electron triggers always demand additional FastCalo processing time.

Figures 9 and 10 show comparisons of the total processing time per call for the feature extraction and hypothesis-testing algorithms in the FastCalo step respectively (see the related FastCalo blocks in Figure 1).

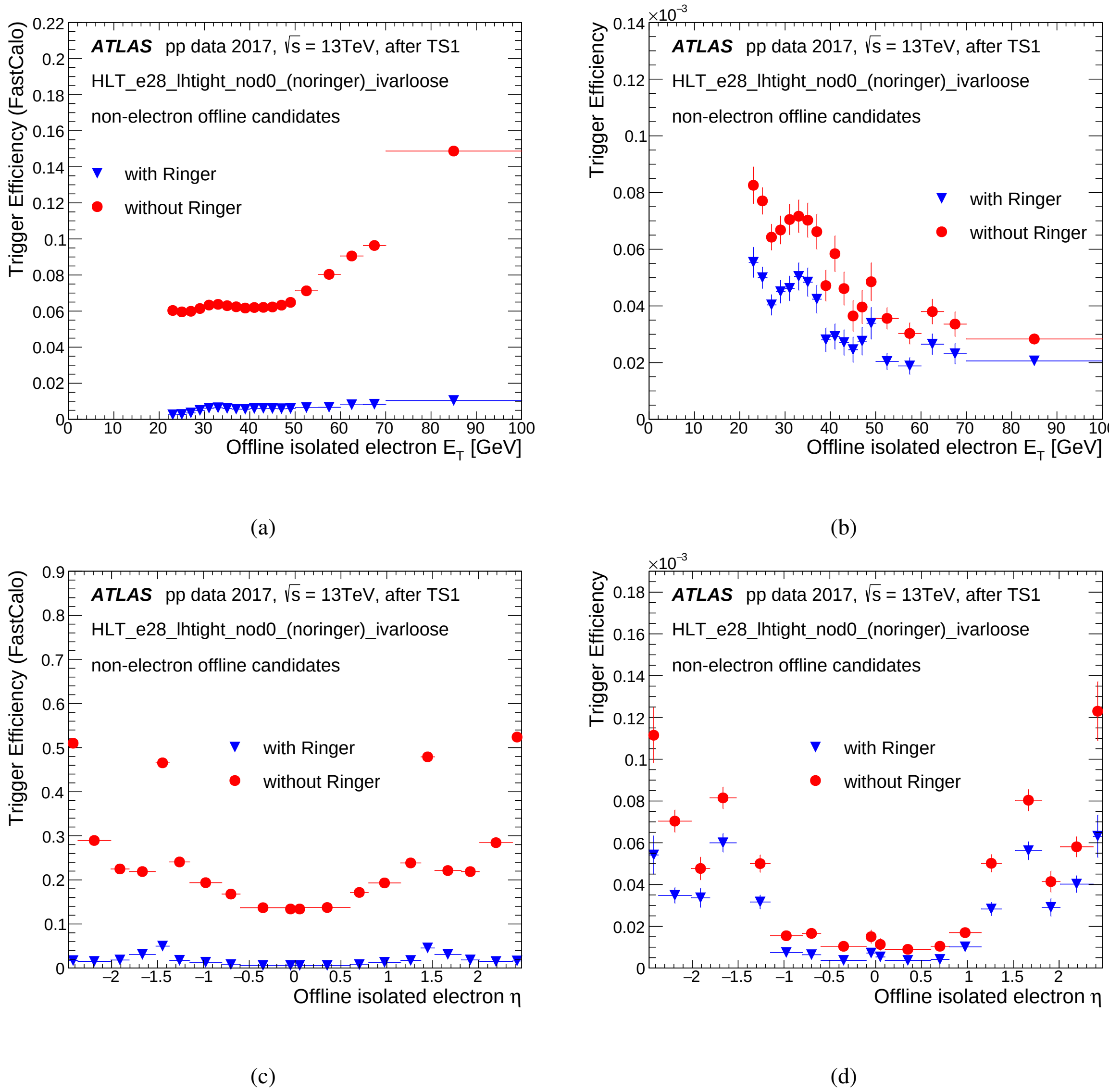


Figure 8: Efficiency of selecting background at the FastCalo (left) and HLT (right) steps as a function of E_{T} (top) and η (bottom) with a single-isolated-electron trigger requiring $E_{\mathrm{T}} > 28$ GeV and 'tight' selection, with and without the Neural-Ringer algorithm, employing 2017 collision data collected after the Neural-Ringer algorithm deployment.

The processing time per call does not include the online data preparation time. The measurements are based on a very loose electron trigger, with individual executions in the same dedicated computer node.

The measurement employed events from the enhanced bias (EB) stream [21] typically extracted from a one-hour acquisition period with a specific trigger menu based only on first-level selections and a goal of collecting about one million background events more likely to be accepted by the HLT. This is achieved by applying increased weighting in the high-p_{T} region, and at an output trigger rate of 300 Hz [22] for one hour. Hence, the measurements are performed under pile-up conditions, with the reconstruction algorithm executing for multiple RoIs in the same bunch-crossing event.

A multi-modal structure can be observed in the feature extraction distributions of both trigger configurations;

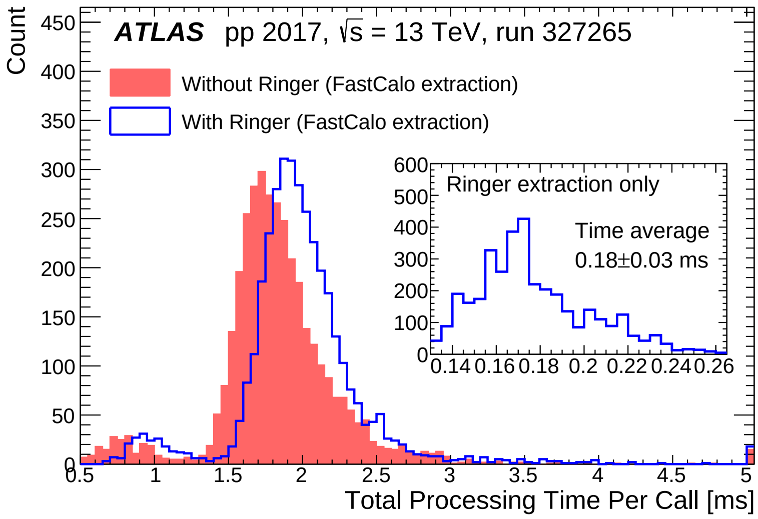


Figure 9: Total processing time per call for the feature extraction algorithms in the FastCalo step of electron triggers rerunning offline, with (blue) and without (red) the Neural-Ringer, on EB events (peak $\langle\mu\rangle = 45$). The inset on the right shows the processing time per call for only the ring variable extraction.

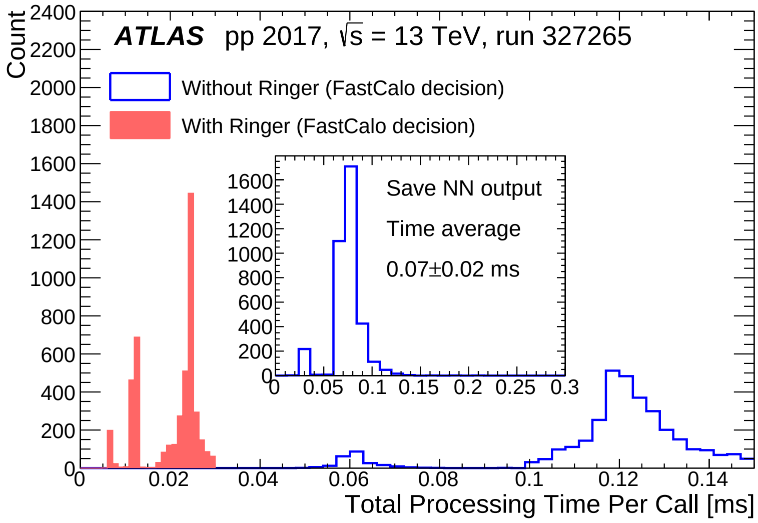


Figure 10: Total processing time per call for the hypothesis-testing algorithms in the FastCalo step of electron triggers rerunning offline, with (blue) and without (red) the Neural-Ringer, on EB events (peak $\langle\mu\rangle = 45$). The centre histogram represents the processing time per call to store the whole Neural-Ringer output in persistent file format.

it is associated with the algorithm for retrieving the EM2 cell energies and building related shower-shape variables.[3] The computation of the ring variables, which is the only difference between the triggers in the FastCalo feature extraction, requires an additional average processing time of 0.18 ± 0.03 ms/event.

[3] The three-peak structure comes from the raw data conversion. This makes one of the largest demands contributing to FastCalo total CPU time.

A smaller contribution comes from hypothesis testing, in which the ring variables are normalized, the discriminating variable's value is computed, compared with the selection requirement, and stored in persistent file format.[4] Specifically, hypothesis testing for the Neural-Ringer is considerably more time-demanding (up to 0.14 ms/event, mostly due to taking 0.07 ± 0.02 ms/event to store the neural-network output) than the cut-based selection (0.02 ms/event), but less time-demanding than feature extraction. Additionally, the time increase is negligible in comparison with the average processing time of a single-electron trigger across the entire HLT (≈ 500 ms in Run 2).

Taking these values into account, the Neural-Ringer can require up to 50% more FastCalo CPU time per event than the trigger without the Neural-Ringer.[5] Nonetheless, the Neural-Ringer provides a more discriminating selection, which reduces the number of CPU-demanding triggers by reducing the need to process fake electrons in subsequent steps that are more computationally demanding. In this measurement, where about 3400 events from the EB stream were considered, the Neural-Ringer algorithm provided an overall 60% reduction in the CPU demands (from 30.72 ms to 10.36 ms per event) by rejecting most fake electron candidates during the FastCalo step.

[4] For Run 3, this step has been removed as a way to reduce the CPU time at the FastCalo step.

[5] The CPU-time increase is expected to depend on the trigger configuration because of the presence of pile-up. The reported results are for the HLT_e17_lhvloose_nod0 single-electron trigger.

4 Neural-Ringer assessment vs cut-based triggers

This section presents a more detailed assessment of the Neural-Ringer's behaviour in the period 2017–2018 relative to the previous electron trigger's cut-based strategy. A $Z \to ee$ tag-and-probe selection was used to evaluate the level of agreement between the two triggers, as well as possible biases due to the variables the Neural-Ringer employs in the likelihood algorithm. Since both the offline and final HLT electron selections are based on such variables, limiting the evaluation to those varaibles suffices for understanding the possible impact of the Neural-Ringer algorithm in most analyses. In addition, it is advantageous to compact the input space into a set of a few such variables that have low correlation and are easy to interpret.

Since each tag-and-probe event has one electron, called the probe electron, that was not used to select that event at trigger level, two offline analyses were performed to assess the trigger performance with and without the Neural-Ringer, with the offline selection applied either to the tag electron (Homogeneity Tests) or to the probe electron (Quadrant Analysis).

The Quadrant Analysis allows a direct assessment of possible bias due to the introduction of the Neural-Ringer by comparing the profiles for all possible disjoint decisions from the two trigger classifiers. It is possible for the classifiers to agree, with both of them accepting or rejecting the event. Likewise, they can disagree in two possible ways, with one classifier accepting the event and the other rejecting it.

Although it is reasonable to assume that the tag and probe showers develop independently, and that the Neural-Ringer when applied to the tag selection cannot, in principle, alter the profile of the probes besides changing the number of tag-and-probe pairs, this assumption should be tested. An evaluation of any possible systematic effect due to using the Neural-Ringer in obtaining the likelihood probability distribution functions (PDFs), used for offline and final HLT selection, is performed with the Homogeneity Tests.

4.1 Quadrant Analysis

The Quadrant Analysis was developed to compare the decisions of two classifiers. Given signal candidates and two classifiers, these are the possibilities: either both accept or reject the candidates, or only one accepts the candidates. The four possible decision quadrants are evaluated from shower-shape distributions. Furthermore, in this analysis, it is also possible to evaluate the disagreement between two classifiers. The disagreement between two classifiers **A** and **B** can be defined as:

$$\text{disagreement} = \frac{N_{\mathbf{A}} + N_{\mathbf{B}}}{N_{\mathbf{A}} + N_{\mathbf{B}} + N_{\mathbf{AB}} + N_{\mathbf{!AB}}} \tag{2}$$

where $N_{\mathbf{A}}$ ($N_{\mathbf{B}}$) is the total number of candidates accepted exclusively by classifier **A** (**B**) and $N_{\mathbf{AB}}$ ($N_{\mathbf{!AB}}$) is the total number of candidates accepted (rejected) by both classifiers.

In this approach, only collision data collected using the primary triggers with an equivalent offline selection are considered.[6] Results in Section 4.1.1 show trigger selection performance using the 'tight' requirement for the duplicated triggers during 2017.

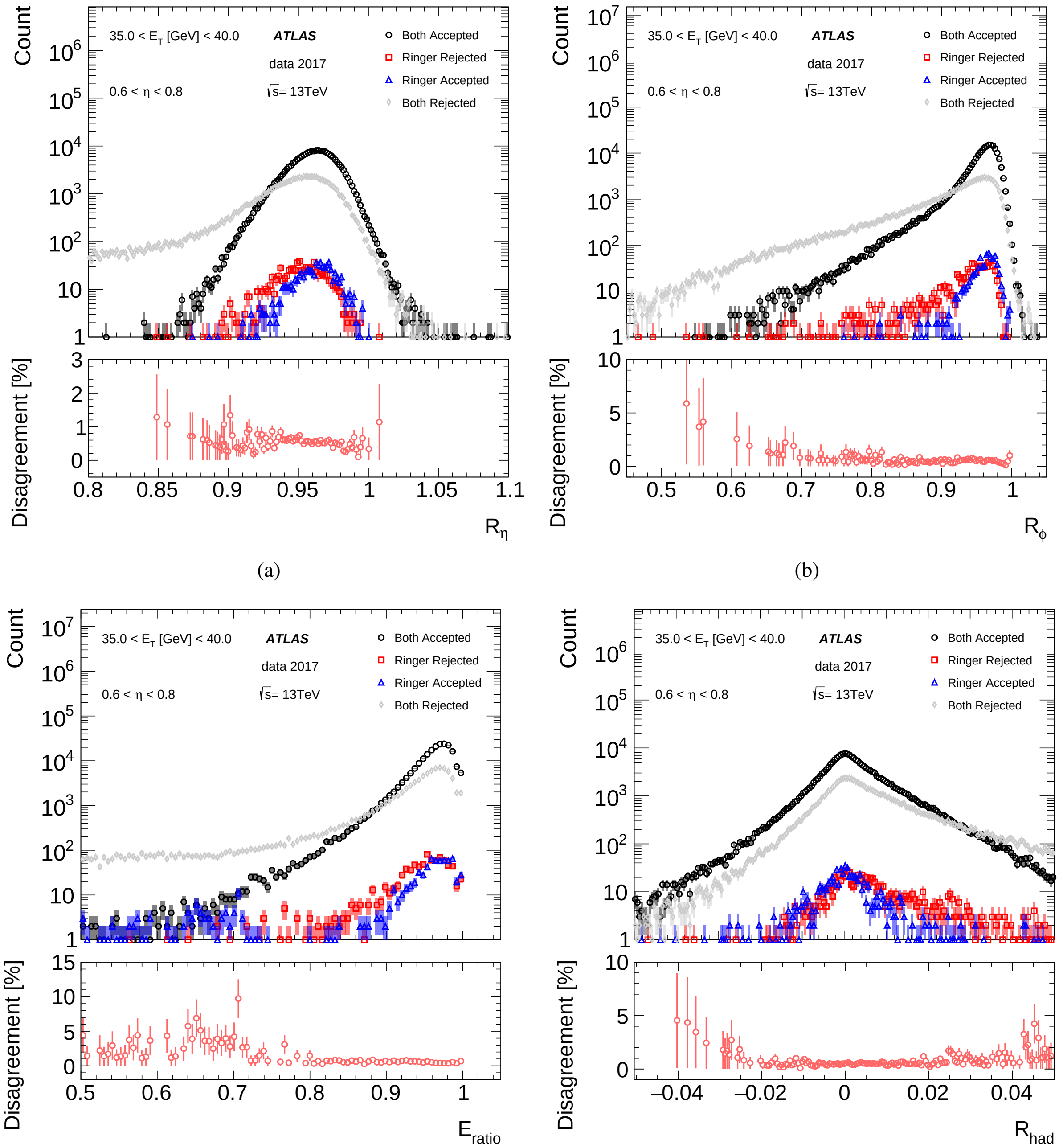
[6] For example, if a 'tight' trigger leg is being evaluated, the 'tight' offline selection is also applied to the candidate.

4.1.1 Results

The Quadrant Analysis was performed on the four variables (R_η, R_ϕ, E_{ratio} and R_{had}) used in the likelihood [2] selection that have the best discrimination power between electrons and jets.

As shown in Figure 11, the disagreement is small and in most cases bounded at about the 1% level for the coverage of each variable. Here, only the results for $0.6 < |\eta| < 0.8$ and $35 < E_{\text{T}} < 40$ GeV are shown, but this behaviour is also observed for other E_{T} regions, and these results are more representative of those obtained in other regions of the $E_{\text{T}} \times |\eta|$ phase space. One should note that the working points of the two triggers are not exactly the same, although the Neural-Ringer aims to achieve the same signal efficiency as the cut-based strategy and was operating as desired. In other words, the height differences between the blue and red profiles in Figure 11 are mainly due to small efficiency differences between the two triggers.

The two triggers behave very similarly for all variables in the region shown in Figure 11, even if some slight shifts between blue and red profiles can be seen. This is the case for R_η and R_ϕ, where the Neural-Ringer trigger consistently selects slightly more events in the signal region, i.e. ones with slightly tighter showers in η and ϕ, respectively, in the EM2 layer. The Neural-Ringer trigger's behaviour shows an even smaller effect in R_{had}, where tighter tails are observed, resulting in fewer electrons with 10% to 30% of their energy in EM1, 1% to 2% in EM3 and more than 4 GeV hadronic leakage. The E_{ratio} variable also shows the Neural-Ringer trigger has a slight bias towards selecting more events in the signal region. Interestingly, a few events with $0.5 < E_{\text{ratio}} < 0.7$ are accepted by the trigger with the Neural-Ringer, probably due to electrons resulting from premature showers with a barycentre near the boundary between two strips. Although similar behaviour is seen in other regions, the differences vary in strength between $|\eta|$ regions.



(a)

(b)

(c)

(d)

Figure 11: Quadrant Analysis plots for the main offline-reconstructed calorimetry variables employed in the likelihood and for the $0.6 < |\eta| < 0.8$ and $30 < E_T < 35$ GeV region. The top pad in each figure shows the raw number of candidates for the four mutually exclusive cases: accepted both with and without the Neural-Ringer (black); accepted only with the Neural-Ringer (blue); accepted only without the Neural-Ringer (red); rejected both with and without the Neural-Ringer (grey). The bottom pad shows the disagreement as defined in Eq. (2).

4.2 Homogeneity Tests

In this analysis, the impact on the likelihood algorithm of using different triggers, Neural-Ringer and cut-based, is evaluated. This study requires a pair of duplicated triggers, applying an $E_{\mathrm{T}} > 28$ GeV isolated and ‘tight’ electron selection with and without the Neural-Ringer, operating together online. The analysis is performed using $Z \to ee$ tag-and-probe collision data, which were selected with the trigger requirement set to either one of the duplicated online triggers for the tag, whereas the probe selection is invariably set to the offline VeryLoose criterion. This is exactly the same set-up that ATLAS employed to derive the offline likelihood for electrons with E_{T} above 15 GeV [2]. To benefit from the full 2017 dataset, collision data collected before TS1 were also employed. If the profiles are statistically identical, the Neural-Ringer trigger is not expected to cause any relevant change in the derivation of the offline likelihood PDFs.

The problem is addressed by applying homogeneity tests to histograms [23] in order to check for systematic effects of the trigger configuration. These tests are based on a test originally developed by Pearson [24] and popularly employed in many fields beyond high-energy physics, e.g. social sciences [25] and health [26], where they are usually called contingency or consistency tests.

In order to benefit from the tests without having to customize them to an ATLAS-specific analysis set-up, the collision data were split into two statistically independent groups, consisting of data alternately taken from consecutive short time periods to keep their data-taking conditions as similar as possible.

The p-value, or probability value, is a measure used in statistical hypothesis testing to determine the significance of one’s results. It represents the probability of obtaining an observed result, or something more extreme, if the null hypothesis (which states that there is no effect or no difference) is true. By comparing the p-values of the two groups, it is possible to evaluate how the different configurations affect the likelihood that the profiles are being drawn from the same distribution. If the p-values have negligible fluctuations with respect to the values obtained from tests comparing the same triggers, then the systematic effect in the profiles is negligible for a dataset that is half[7] of the size of the whole available dataset.

To provide better insight into possible distortions, the corresponding individual χ contributions of each group are computed, allowing them to assume positive or negative values by defining

$$\chi^{s}_{i,j} = \frac{(r_{i,j} - b_{i,j})}{\sqrt{b_{i,j}}}, \tag{3}$$

where $r_{i,j}$ ($b_{i,j}$) is the number of observations collected by the trigger with (without) the Neural-Ringer in j^{th} histogram bin and in the i^{th} $E_{\mathrm{T}} \times |\eta|$ region.

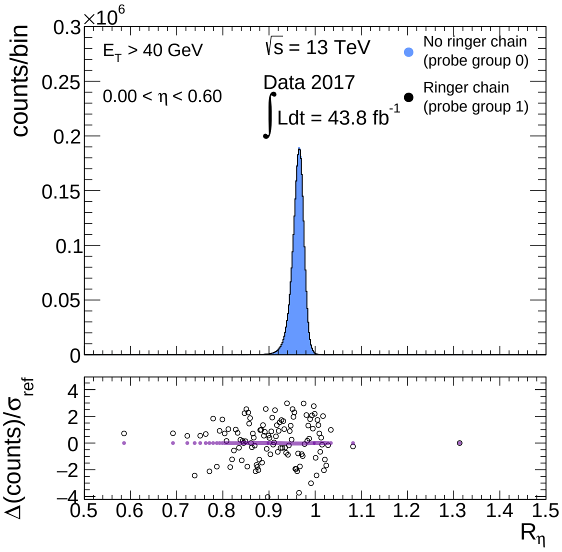
4.2.1 Results

Regardless the trigger configuration, the homogeneity tests between the two groups result in similar p-values. In other words, the fluctuations in the profiles are mainly statistical, with no sign of systematic effects due to the trigger configuration.

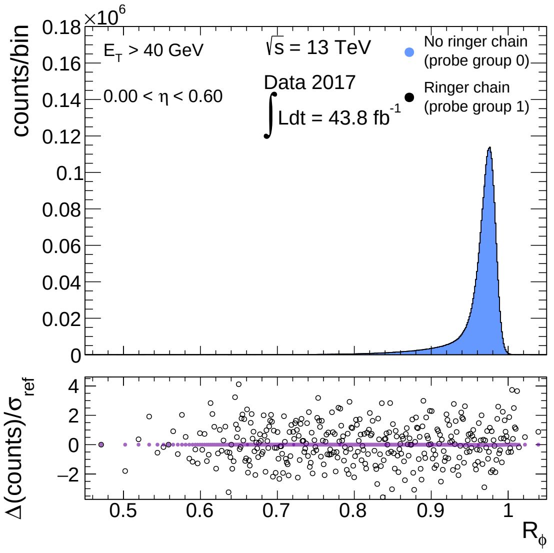
Figure 12 shows the residuals for the R_{η}, R_{ϕ}, E_{ratio}, and R_{had} calorimetry variables when the reference histogram is obtained from collision data collected by the trigger without the Neural-Ringer for the first

[7] Each group contains approximately half of the integrated luminosity.

group. The residuals are dominated by statistical fluctuations, and are within the expectations from the homogeneity hypothesis for most variables and regions. Similar behaviour is observed for track variables.



(a)



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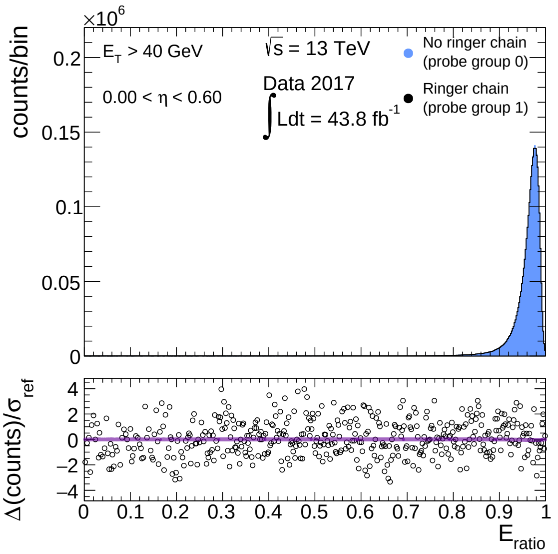

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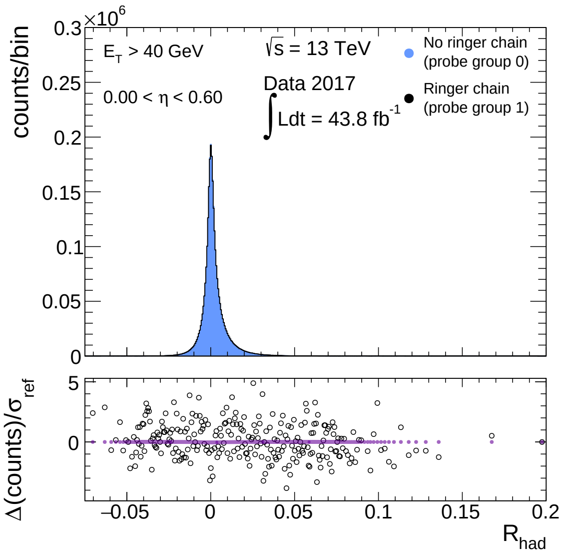

(d)

Figure 12: Top: Profile histograms for the R_η, R_ϕ, E_{ratio}, and R_{had} calorimetry variables employed in the offline likelihood in the $E_{\text{T}} > 40\,\text{GeV}$ and $0.0 < |\eta| < 0.8$ regions using collision data collected by triggering without the Neural-Ringer in the first arbitrary data group (blue area histogram) and collision data collected by triggering with the Neural-Ringer in the second arbitrary data group (black line histogram). Bottom: residual contributions using the χ^s statistic (black circles, from Eq. (3)), and the expected χ^s residuals if there are no distortions or statistical fluctuations with respect to the reference (purple dots at zero).

5 Conclusion

The Neural-Ringer has an innovative design for electron selection based on calorimetry information. With the goal of satisfying specific trigger system demands, feature extraction from concentric ring energy sums provides an efficient description, which is quickly derived through simple operations. Neural-network models are employed to exploit non-linear correlations between rings, and the optimization of this strategy culminated in the deployment of a set of models specific to each region of pseudorapidity and transverse energy.

In the second half of 2017, the Neural-Ringer algorithm replaced a cut-based strategy in the FastCalo decision step of the HLT and became the baseline algorithm for triggering events containing electrons with E_{T} above 15 GeV, as part of the ATLAS online system's CPU-time reduction campaign. Furthermore, it was the first time a neural-network method was employed as a baseline event-selection algorithm in the ATLAS trigger system.

During 2017 operation, with the Neural-Ringer algorithm trained on simulated data, fake-electron trigger rates were reduced by a factor of 13 in the FastCalo step and by a factor of 2 overall in the HLT with respect to the offline likelihood algorithm, while achieving high electron-selection efficiencies very similar to those of relevant single-electron triggers without the Neural-Ringer. For 2018, the Neural-Ringer was trained with collision data, resulting in an estimated additional reduction factor of 1.19–1.69 for fake-electron triggers in the FastCalo step with respect to 2017. The Neural-Ringer was also studied from an offline likelihood perspective to obtain additional insights into its behaviour by using two equivalent single-electron triggers with and without the Neural-Ringer. The 'quadrant analysis' showed a high level of agreement between the two triggers, with the Neural-Ringer trigger showing a slight bias towards the signal region in the disagreement profiles of some calorimetry-based variables. Finally, the 'homogeneity tests' between the two groups of data resulted in similar p-values. In other words, the fluctuations in the profiles are mainly statistical, with no sign of systematic effects due to the trigger configuration.

Physics beyond the Standard Model is getting more attention at the LHC and is expected to be part of the main focus in the data-taking phases to come. At the beginning of Run 3, the Neural-Ringer was updated to include events in which electrons have smaller angular separations (such as in boosted topologies), particularly limiting the effect of other contributions at the edge of the feature extraction window, which can result from these physics processes. In addition, the Neural-Ringer was extended into the E_{T} region below 15 GeV, motivated by the high CPU demands of triggers in this kinematic region.

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